Timster's Audio Works - **Split-Flap Clock**



for Dummies*

The unofficial, slightly cheeky passenger's guide to the desk clock that clacks like a 1970s airport.

For Split-Flap Clock v1.1.0

* Dummies not included. Clatter extremely included. Not affiliated with any famous yellow book series, any airline, any airport, or the Italian company whose departure boards taught a generation of travellers to look up and listen. We just really like them.

DEPARTURES

Or, what's in this book. All gates are a short walk from here.

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YOUR BOARDING PASS

Keep this handy. Tear along the dotted line, metaphorically.

HANDS THAT MATTER

Drag the face	Move the clock anywhere
Right-click	The menu: Settings, Keep on top, Hide title bar, Show on all desktops, Sound, Replace the dead bulb (if one's out), Refresh the board, Quit
Double-click	Opens Settings
⚙ top right	Also opens Settings (it appears when the pointer is over the clock)
Resize the window	The board scales to fit and keeps its shape
Alarm ringing	Press the **SNOOZE** or **STOP** cards on the board, or Esc to stop
Click a dark lamp	Puts in a new bulb, when bulbs are replaced by hand

WHERE THINGS LIVE

Yellow or white	Settings, **Digits**
12 or 24 hour	Settings, **Hours**
More places	Settings, **Add a place**
Groups	A place's **Region**
The clatter	Settings, **Refresh** and **Sound**
Wake-up calls	Settings, **Alarms**
Light or dark	Settings, **Board** and **Menus**
Faster or slower	Settings, **Flip speed**
Lamps and bulbs	Settings, **Lighting**
Easier to read	Settings, **Card brightness**

PANIC BUTTONS

Too loud	Right-click, untick **Sound**. Then apologise to the open-plan office.
Can't find the window edges	Right-click, untick **Hide title bar**
Clock too big or small	Settings, **Size**
Want the show right now	Right-click, **Refresh the board**
Alarm won't stop	Esc, or right-click, **Stop alarm**

ANNOUNCEMENTS USED IN THIS BOOK

Listen out for these. They are, unlike the real thing, perfectly audible.

PASSENGER INFORMATION Something that makes life easier, or looks better, or both.

PLEASE NOTE Worth writing on the back of your hand. Or your boarding pass.

SECURITY ANNOUNCEMENT Your settings, your colleagues' patience or your sanity are in danger.

CREW ONLY Nerd corner. Skip it freely; the board will keep clacking without you.

GATE 01 WHAT ON EARTH IS A SPLIT-FLAP DISPLAY?

Picture an airport in 1975. Wide lapels, brown carpet, somebody smoking under a sign that says not to, and high above the check-in desks a long black board. Every few minutes the whole thing erupts: hundreds of little cards whirring and clattering at once, letters tumbling, until the noise dies away and there it is, settled and calm. **LONDON 14:35 BOARDING**. Every head in the hall turns up to look.

That was a **split-flap display**. Each character on it is a small drum of hinged cards, a little like a Rolodex. Every card carries the top half of one symbol on its front and the bottom half of the next on its back. Let a card fall and the next symbol appears, top and bottom meeting at the split. To get from A to F, the drum has to flip through B, C, D and E on the way. There's no going backwards. That's why the board spins and why it clatters.

The best known came from an Italian company in Udine, Solari, from the 1950s on, which is why travellers and designers still call them **Solari boards**. The architect Gino Valle, who designed for them, also drew a split-flap desk clock in the 1960s, the Cifra 3, which ended up in New York's Museum of Modern Art. By the 1970s and 80s split-flap boards were in airports and railway stations all over the world. Screens replaced most of them from the 1990s on. Screens are cheaper and easier to update, but they're silent and nobody misses them.

Split-Flap Clock is a love letter to those boards. It doesn't play a video of flaps. Every card is a real little drum in the program: it knows which symbols it carries and in what order, and it can only move forward. Every fall is drawn as it happens, with the top half folding down and the bottom half swinging into place.

> PLEASE NOTE Split-Flap Clock is a desk clock, not a timetable. It shows your time, the date, and as many other places as you like, grouped into regions, in the colours you choose. It does not know when your flight leaves. Sorry.

> CREW ONLY The minute's tens card has a drum of only six flaps, 0 to 5, just like a real flap clock. So when 10:59 becomes 11:00, both minute cards each fall exactly once: 5 to 0 and 9 to 0. The hour's tens card is 0 to 2 on a 24-hour clock, and blank or 1 on a 12-hour one. Names and dates use the full alphabet drum, which is why they take longest to arrive.

GATE 02 UP AND CLACKING IN UNDER A MINUTE

1. **Collect your ticket.** On the clock's GitHub page, under **Releases**, download `splitflap-clock.exe` for Windows, or `splitflap-clock-linux-x86_64` for Linux.

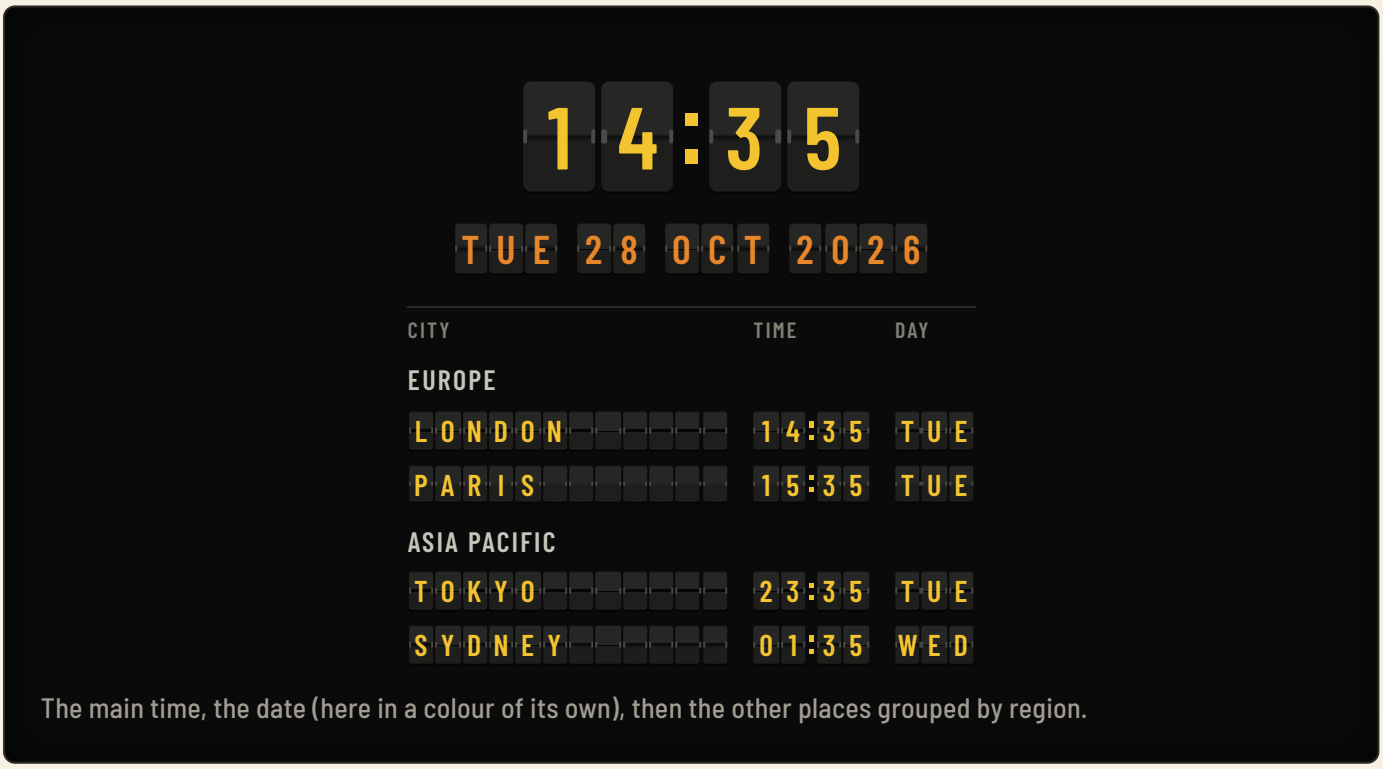
2. **Check in.** There's nothing to install. On Windows, double-click the exe. On Linux, make the file executable once (`chmod +x splitflap-clock-linux-x86_64`, or tick "Allow executing" in its properties) and then run it.
3. **Watch the board come in.** Every card starts blank, then they all set off at once, a moment apart and each at its own speed, until the time and the date settle into place.
4. **Look up.** Feel faintly nostalgic for a decade you may not have lived through. That's normal.

> SECURITY ANNOUNCEMENT Windows may say it "protected your PC" from an unrecognised app the first time. The clock isn't signed by a big company, so SmartScreen hasn't heard of it. Click **More info**, then **Run anyway**. It only asks once.

> PASSENGER INFORMATION The clock is a single file. Put it somewhere you'll keep it, such as a Programs or Apps folder, rather than leaving it in Downloads, especially if you're going to have it start when you log in (Gate 14).

GATE 03 A GUIDED TOUR OF THE BOARD

Here's a board with everything switched on. Yours starts simpler: the time and the date, in yellow.



The main time, the date (here in a colour of its own), then the other places grouped by region.

The main time

Your computer's own time, in big cards: hours and minutes, and seconds if you ask for them. In 12-hour mode the hour's tens card goes blank before 10 o'clock, as on the real thing, and a small **AM** or **PM** card sits beside the minutes.

The date

The day, date, month and year, as in `TUE 28 OCT 2026`. You can switch it off.

Other places

As many as you like, listed under the columns **CITY**, **TIME** and **DAY**. Each has a name of up to 12 cards, its time and its day of the week. The day is how you can tell that it's already tomorrow in Sydney.

The sky

If you switch on **Day and night**, a card at the end of each place's row, and one after the main time, shows the sky there: morning, day, evening or night. See Gate 09.

The lamps

If you choose a **Lighting** mood, a row of warm lamps hangs over the top of the board, and the cards stand out in their light. See Gate 10.

Region headings

If you give places a region, they're grouped under a heading for it, in the order you choose. Places without one go under **OTHER**, unless they come first on the board. See Gate 07.

> `CREW ONLY` The name cards carry the letters A to Z, the digits and a little punctuation: `. - + : / ' & , ( )`. Names are shown in capitals. Anything else, an `é` for instance, still appears, but it arrives in a single flip, because there's no such flap on the drum to come round to.

`GATE 04` DRIVING IT, OR WHERE ALL THE BUTTONS WENT

There are no buttons on the board itself. A real board doesn't have any either. Everything is done by hand:

Moving it

Press and drag anywhere on the face. That works with or without a title bar.

Sizing it

Drag the window's edge, and the board scales to fit while keeping its shape. Or use **Size** in Settings, which goes from 0.30 to 4.00 times the normal size. When you add places, the window grows to fit them, but never taller than your screen: past that, the whole board shrinks instead.

The menu

Right-click the face for **Settings...**, **Keep on top**, **Hide title bar**, **Show on all desktops**, **Sound**, **Refresh the board** and **Quit**. When an alarm is ringing it has **Snooze 9 minutes** and **Stop alarm** too, and when a bulb is out and waiting for you, **Replace the dead bulb**.

Settings

Double-click the face, choose it from the menu, or click the small ⚙ that appears in the top right corner when the pointer is over the clock. Settings opens in its own window, right beside the clock: to its right, or to its left if the clock is near the edge of the screen. Changes take effect straight away; there's no OK button to forget. At the foot of Settings it says which version you have and when it came out, next to **Beer money** (see the Acknowledgements).

> PASSENGER INFORMATION **Hide title bar** turns the clock into a bare black slab that sits on your desktop like the real thing. You can still drag it about by its face. To make it bigger or smaller without the window edges, use **Size**.

> PLEASE NOTE Settings are saved when the clock closes, and loaded next time it starts. They live in `%APPDATA%\splitflap-clock` on Windows and `~/.local/share/splitflap-clock` on Linux. Updating to a new version keeps them.

GATE 05 THE DISPLAY SETTINGS, SWITCH BY SWITCH

The top of the Settings window, row by row. Below it come **Alarms** (Gate 13), **Add a place** and **Other places** (Gate 06), and **Regions** (Gate 07).

ROW	WHAT IT DOES
Digits	**Yellow**, the classic airport amber-yellow, or **White**, the railway-station look. Everything without its own colour uses this. (White cards always have dark letters.)
Board	**Black cards**, as on most boards, or **White cards** with dark letters, as on some stations' boards. See Gate 08.
Lighting	The lamps over the board: **Off**, **Soft lamps**, **Warm lamps** or **Late evening**. Then how often bulbs fail, and who replaces them. See Gate 10.
Card brightness	Lighter cards with brighter pins, so the split and the edges show up better. **Classic** goes back. See Gate 08.
Menus	The right-click menu and the Settings window: **Like my system**, **Light** or **Dark**.
Colours	Own colours for the main **Time** and the **Date**. See Gate 08.

Hours	**12-hour** or **24-hour**. It applies to every place on the board.
Show	**Seconds** adds two more big cards to the main time. **Date** shows or hides the date line.
Day and night	A sky card for each of the **Places**, and for the **Main time**. See Gate 09.
Flip speed	How fast the flaps fall, from 0.25x to 4x, with **Normal** to go back. See Gate 11.
Refresh	How often the whole board does its clatter: **Never**, **Every 15 minutes**, **Every 30 minutes**, **Every hour** or **At random**. See Gate 11.
Sound	The clatter you hear, its **Volume**, and whether the time ticking over makes a sound too. See Gate 12.
Window	**Keep on top** of other windows, **Hide title bar**, and **Show on all desktops**. See Gate 14.
Start	**Start when I log in**. See Gate 14.
Size	How big the board is, from 0.30 to 4.00 times the normal size.

> PASSENGER INFORMATION Seconds look wonderful but turn the clock into a busy little machine, flipping every second. For a calm desk, leave them off and let the minutes do the work.

GATE 06 OTHER PLACES, OR THE ARRIVALS HALL

Below Display are **Add a place** and **Other places**. You can have as many as you like: the board shrinks to fit when they run out of screen.

Adding a place

1. In **Add a place**, type a city or region into the search box, such as `Tokyo`, `New York` or `Europe/`. The clock knows every time zone in the world by its official name, such as `America/New_York`, and a list of matches appears.
2. Click the one you mean, or don't: the best match is picked already.
3. Press **Add** (it says which, e.g. **Add TOKYO**) or press Enter. The new place appears at the bottom of the list and flips onto the board.

> PASSENGER INFORMATION Time zones are named after a big city in them, not necessarily yours. Manchester runs on `Europe/London`, Melbourne doesn't have to be `Australia/Sydney`, and most of the United States is covered by a handful of names. Pick the zone, then rename the place to whatever you like.

Each place

Every place has its own panel in **Other places**:

The name

What the board says, up to 12 characters. It starts as the city from the time zone's name, which you can change: **MUM**, **HQ** and **BASE CAMP** are all fine.

▲ ▼ and Remove

Move the place up or down the list, or take it off the board.

Region

Which group it goes in on the board, up to 24 characters. See Gate 07.

Colour

Its own colour. See Gate 08.

Time

From its time zone, the usual, which follows summer time by itself, or **Set by hand**.

Setting a time by hand

Sometimes the time zone isn't the time you want: a ship, a research station, a friend whose clocks run on their own rules, or a place that doesn't match any zone. Choose **Set by hand**. Two boxes appear showing the time there now, hours and minutes. Change them to the time it should be, by typing or dragging, and from then on the clock keeps that place exactly that far ahead of or behind UTC. Next to the boxes it says how far, as in `UTC+05:45, no summer time`.

> PLEASE NOTE A time set by hand stays at that offset all year: it does **not** change for summer time. If the place does, either change it by hand twice a year, or switch back to **From its time zone**.

> CREW ONLY Any clock time can be either side of midnight, as in 23:00 today at UTC+13 or 23:00 yesterday at UTC-11. The clock keeps whichever of the two is nearer to where the place was before you changed it. So nudging the time doesn't flip the day, and the **DAY** column stays right.

GATE 07 REGIONS, OR SORTING YOUR LUGGAGE

With a dozen places on the board, it helps to group them. Type a region into a place's **Region** box, such as Europe, Clients, Family or Night shift, and give the same region to the places that belong together. On the board they're gathered under a heading for it.

- Regions are matched without fuss over capitals or stray spaces: Europe, EUROPE and europe are the same region. Headings are shown in capitals.
- Within a region, places keep their order from the list in Settings.
- Places with no region form a group of their own. If that group is first on the board, it sits straight under the column captions. Anywhere else it gets the heading **OTHER**, so it doesn't look like part of the region above.

Putting regions in order

As soon as any place has a region, a **Regions** list appears in Settings. It shows every region in its order down the board, including *Places with no region*. Use ▲ and ▼ to move whole groups up and down. New regions join the bottom of the list.

> PASSENGER INFORMATION Put the region you look at most at the top, just under the date, where your eye lands first.

GATE 08 COLOURS, OR THE PAINT SHOP

Real boards came in one colour, usually yellow or white on black. This one doesn't have to. The main **Time** and the **Date** (in Display), every place (in its own panel) and every alarm (in its panel, Gate 13) have a colour of their own. Each has three parts:

The colour button
: Click it for the colour picker: drag around the colour square, or type the red, green and blue values into the boxes underneath.

The colour box
: Next to the button, showing the colour as a code like #F4C430. Type or paste a colour into it in any of the usual ways: #F4C430, F4C430, the short #FC3, or 244, 196, 48. The colour changes as soon as what's in the box is a colour; until then it stays as it was.

Reset
: Appears once something has its own colour, and takes it back to the board's yellow or white.

> PASSENGER INFORMATION To give several places exactly the same colour, click the 📋 button in one place's colour picker. It copies the colour as R, G, B. Then paste that into the colour box of each of the others. You can also select the code in a colour box and copy it with Ctrl+C.

PLEASE NOTE Anything without its own colour follows **Digits**. So you can switch the whole board between yellow and white and keep just your favourite places in their own colours.

White cards

Not every board was black. Set **Board** to **White cards** for light cards with dark letters on a pale grey board, as some railway stations had. Your own colours still apply on white cards, so pick ones that show up on a light card.

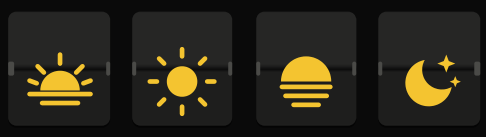
Card brightness

A real board's cards were black on black, which is lovely from across a hall and hard work up close. **Card brightness** lifts them: slide it up and the cards turn a lighter charcoal and their hinge pins brighten, so the split across the middle and the edges of each card stand out. On white cards it works the other way round, with a darker split and darker pins. **Classic** takes it back to the usual look.

SECURITY ANNOUNCEMENT Dark colours on black cards are a fine way to hide a city from yourself. If a place seems to have vanished, check its colour.

GATE 09 DAY AND NIGHT, OR THE VIEW FROM THE WINDOW

A clock tells you the time in Tokyo. It doesn't tell you whether Tokyo is asleep. Switch on **Day and night** in Settings, for the **Places**, the **Main time** or both, and each gets one more card: a picture of the sky there now.



Morning, day, evening and night, in the order the card turns.

CARD	WHEN
Morning, the sun rising over the horizon	The hour around sunrise: from half an hour before to half an hour after.
Day, the sun	From then until the hour around sunset.

Evening, the sun sinking into the sea	The hour around sunset: half an hour either side.
Night, the moon and stars	When the sun is down.

The card's flaps go round in the order of a day, so each change is a single flip. On a refresh it spins through all four.

Sunrise and sunset are worked out for each place, for each day, from where its time zone's city is. That's built into the clock, so nothing is looked up online. So the night really is most of the day in Reykjavik in December, and summer at Svalbard never gets dark at all. The main time's card uses the city of your computer's own time zone.

> PLEASE NOTE A place whose time is **set by hand**, or a zone with no city, such as UTC, has no location to go by, so its sky goes by its clock: sunrise at six in the morning and sunset at six in the evening.

> CREW ONLY The sums are the usual ones for the sun's position, good to a minute or so. A time zone's city stands for the whole zone, so for somewhere far from it, the card can change a little before or after the sun really does.

GATE 10 LIGHTING, OR THE LAMPS OVER THE HALL

Departure halls were never evenly lit. The boards hung under rows of warm incandescent lamps, each throwing its own pool of light, so the cards near a lamp glowed and the ones between them sat in shadow. And every so often a bulb would give up, and a stretch of the board went dark until somebody came with a ladder. Choose a **Lighting** mood in Settings and the clock does the same.



Late evening, with the right-hand bulb gone. Somebody fetch the ladder.

LIGHTING	THE LOOK
Off	No lamps: the board evenly lit, as it always was. This is how the clock comes.
Soft lamps	A gentle warm wash, the lamps only just showing.
Warm lamps	A clear pool of light under each lamp, fading down the board.
Late evening	The hall lights down, the lamps doing all the work: dimmer, more amber, with deep shadows.

The lamps hang just over the top edge, one for every few cards' width, so a wide board gets more of them. Each card throws a soft shadow down onto the board behind it, deeper the dimmer the mood. It's what makes the board look like a thing hanging on a wall rather than a picture of one.

Bulbs

Next to the mood, choose how often bulbs fail: **Fail every few days** (the usual), **Fail about once a day**, or **Never fail**. When is left to chance, so a bulb might go minutes after the last one, or not for a week. Only the time the clock is running counts.

A failing bulb flickers and stutters for a few seconds. Most of the time it steadies and carries on. Sometimes it dies, and its pool of the board goes dark.

Replacing them

Replaced for you

The usual. Someone comes round with a new bulb after a minute or so, and it sputters back on.

Replace by hand

A dead bulb stays out until you do something about it, even if you close the clock and start it again. Move the pointer over the dark lamp at the top of the board: it turns into a hand and says **Click to put in a new bulb**. Click, and the new one sputters and comes on. Or right-click the clock and choose **Replace the dead bulb** (or **Replace the 2 dead bulbs**, if you've let things slide).

> PASSENGER INFORMATION **Late evening** with **Card brightness** turned up is the easiest dim board to read: the lamps set the mood and the lighter cards keep the letters crisp.

> PLEASE NOTE The lighting is only a look. A dark patch never hides the time: the cards under a dead bulb still flip, just in shadow. If the whole board looks dim, choose **Off**.

CREW ONLY The bulbs behave like filaments: they take a few hundredths of a second to warm up and cool down, which is why a flicker looks like a stutter rather than a switch. The light is drawn over the finished board, so it falls on the cards, their letters and the board alike.

GATE 11 REFRESHES, OR THE GREAT CLATTER

Every so often, a real board would refresh: every card spinning round its drum at once, and then everything landing back where it was. Split-Flap Clock does the same. On a refresh, every card makes one full turn of its drum, through every symbol it carries, and lands back on the right one.

Like a real board, it's not tidy. The cards all start a moment apart and each runs at its own speed, so the board scrambles all over and settles card by card. The same happens when the clock starts.

REFRESH	WHEN
Never	Only when you ask (see below).
Every 15 minutes	On the quarter-hours by your clock: :00, :15, :30, :45.
Every 30 minutes	On the hour and the half-hour.
Every hour	On the hour. This is how the clock comes.
At random	After a random wait, between two times you choose: **Between** 10 **and** 60 **minutes apart** to start with, anything from 1 to 720. Each wait is drawn afresh, so you never know quite when it's coming.

Flip speed

Flip speed in Settings sets how fast the flaps fall, from a quarter as fast as usual, for a stately board, to four times as fast, for a frantic one. It sets everything that moves: each flap's fall, the spin through the alphabet, and the ripple across the board. **Normal** takes it back to the usual speed.

PASSENGER INFORMATION Want one now? Right-click and choose **Refresh the board**. It's the most satisfying thing in this book.

CREW ONLY The quick flips on the way take about 0.07 seconds each, and the last one, landing on the right symbol, takes 0.3 seconds. A name card has 47 flaps on its drum, so a full turn takes it three seconds or so. The digit cards, with only ten flaps or fewer, are done in a moment. That's why the names are always last to settle, on a real board too.

GATE 12 SOUND, OR WHY THE OFFICE HAS GONE QUIET

A split-flap board without the noise is only half the experience. Tick **Clatter when the board starts and refreshes** (or right-click and tick **Sound**) and every flap that falls makes its click. There are no recordings: each click is made as it plays, so no two are quite the same.

- Big cards sound deeper and heavier than small ones.
- Each card is heard from its place on the board, left to right, if you have stereo speakers.
- The last flap of each card lands with a slightly firmer knock.
- A refresh builds to a proper clatter, then thins out to the last few stragglers.

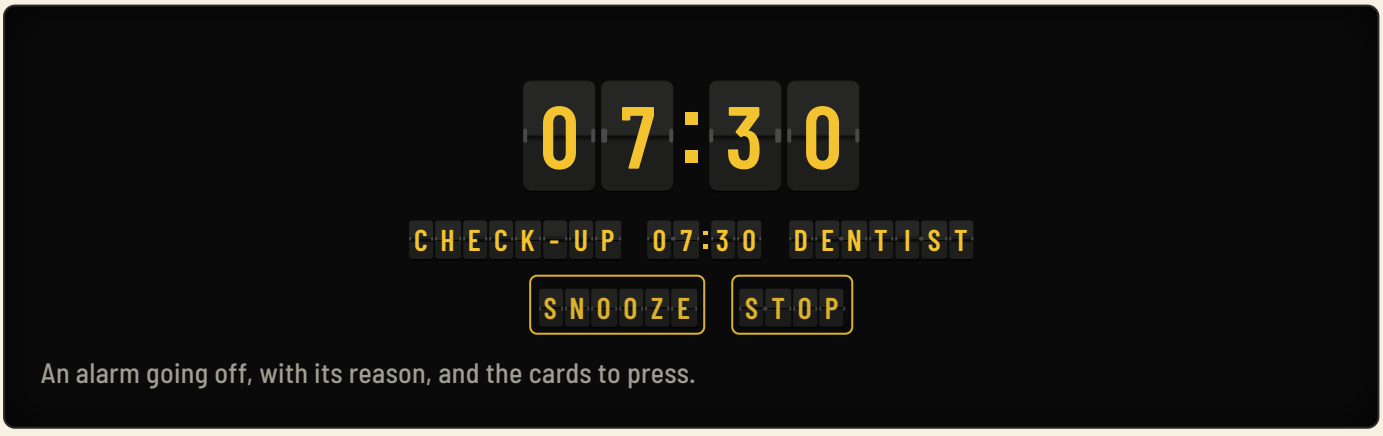
Volume sets how loud. The clock itself is quiet: only the start-up and refreshes make a sound, never the minutes ticking over, unless you also tick **Also when the time changes**.

> SECURITY ANNOUNCEMENT **Also when the time changes**, with **Seconds** on, makes a sound every single second. Some people find that soothing. Their colleagues may not.

> PLEASE NOTE Sound is off when the clock comes. If there's no sound output to play on, Settings says so in orange, and the clock carries on silently. It keeps trying every 15 seconds, so if your speakers come back, or the sound system wakes up after you log in, the clatter returns by itself.

GATE 13 ALARMS, OR YOUR WAKE-UP CALL

An airport never lets a flight leave without telling you. Neither does Split-Flap Clock. Set an alarm and, when it's time, the board announces it: a clatter, a row of its own flipping in under the time, and a proper airport chime, *bing-bong*, every few seconds until you answer it.



An alarm going off, with its reason, and the cards to press.

Setting one

In Settings, under **Alarms**, press **Add an alarm**. Each alarm has a little panel:

The tick, the time and the label

The tick turns it on or off. The two boxes are the hour and the minute, which you can type into or drag. The label is the alarm's name on the board when it rings, up to 12 characters: **WAKE UP**, **TEA**, **CALL MUM**.

Reason

What it's for, such as **DENTIST** or **PICK UP KIDS**, also up to 12 characters. When it rings, the end of its row says the reason. With no reason it says **NOW**.

Colour

Its own colour, next to the reason, so you can tell the dentist from the school run at a glance. When it rings, its row and its **SNOOZE** and **STOP** cards are in that colour. **Reset** goes back to the board's. See Gate 08 for how the colour box works.

Repeat

Click the days it rings on: **M T W T F S S**. The days you've picked light up yellow, and beside them it says what you've picked, such as **Weekdays**, **Weekends** or **Every day**. With no days picked it says **Once**: the alarm rings the next time its time comes round, then turns itself off.

On a date

Tick it to ring once on a particular day instead, such as an appointment next month: set the day, the month and the year. It says **Once, on** and the date. Afterwards it turns itself off.

Time at

Your time, or any place on the board. An alarm at 09:00 at TOKYO rings when it's nine in the morning in Tokyo, summer time and all. It keeps following that place if you move it up or down the list. If you remove the place, the alarm goes back to your time.

Test

Rings it straight away, so you can hear what you're in for.

Remove

Gets rid of it.

Under the alarms, **Chime volume** sets how loud the chime is. The chime has its own volume, and it rings even if the clatter sound is off, because an alarm you can't hear isn't much use.

When it goes off

- The board does a refresh, with its clatter if **Sound** is on, and a row flips in under the time with the alarm's label, its time and its reason, or **NOW**. Under it come two cards to press, **SNOOZE** and **STOP**, which light up under the pointer.
- The chime plays every four seconds.
- **SNOOZE** (or a click anywhere else on the board) makes it go quiet for nine minutes. The row says **DELAYED** with the time it will ring again. Then it all starts again.

- **STOP** stops it. So does Esc, or right-clicking and choosing **Stop alarm**. The right-click menu has **Snooze 9 minutes** too.

PLEASE NOTE The clock has to be running for its alarms to ring. It can't wake a computer that's asleep or switched off. An alarm only rings for times that come round while the clock is running. Starting the clock just after an alarm's time doesn't ring it, and neither does setting or changing an alarm for a time that's just gone. If the computer was asleep when an alarm was due, it still rings when it wakes, as long as that's within 10 minutes. And an alarm that nobody answers gives up by itself after half an hour, so you won't come back from holiday to a clock that's been chiming for a fortnight.

PASSENGER INFORMATION Want to hear an alarm now? Press **Test**. Setting one for the minute it already is won't ring until that time comes round again. Changed an alarm's time after it already rang today? It can ring again today, at its new time.

CREW ONLY Why nine minutes? Because that's what every snooze button has done since the 1950s, and who are we to argue. The chime is two bell notes, E then C, made from a few pure tones as it plays, like the clatter. If two alarms go off together, the board shows the first, and a snooze or a stop answers both.

GATE 14 LOGIN, DESKTOPS AND OTHER GROUND HANDLING

Start when I log in

Tick it in Settings and the clock starts with you every time you log in. On Windows it appears under Startup apps in Task Manager, where you can also switch it off. On Linux it adds an entry to `~/.config/autostart`. If you move the clock's file to another folder, the entry is updated the next time you run it from there.

Show on all desktops

If you use more than one virtual desktop or workspace, this keeps the clock on every one of them, so it's there wherever you switch to.

- **Windows 10 and 11:** the clock has no taskbar button while this is on, and doesn't appear in Alt+Tab. That's the price Windows asks for being on every desktop. Right-click the clock to get at it.
- **Linux:** it asks the desktop to keep the clock on every workspace, in the standard way that desktops with workspaces understand.

Keep on top

Keeps the clock above other windows, like a board above a busy hall.

> CREW ONLY On Linux the clock runs as an X11 window wherever it can, through XWayland on Wayland desktops. Wayland doesn't let apps keep themselves on top or on every workspace, but X11 windows can. The only side effect is that with fractional display scaling, such as 125%, the clock can look a touch soft.

GATE 15 ITINERARIES (RECIPES)

TERMINAL 1, 1975

Digits **Yellow**, **24-hour**, Date on, Refresh **Every hour**, Sound on at a modest volume. Add three far-off places and give them all the region **Departures**. Make a cup of tea at five to the hour and come back for the show.

THE RAILWAY CONCOURSE

Digits **White**, **24-hour**, Seconds on, Refresh **Every 15 minutes**. Hide the title bar and stretch the clock long and low across the top of your screen with **Size**. Mind the gap.

THE WORLD DESK

Add every office you deal with. Give them regions such as **Americas**, **Europe** and **Asia Pacific**, and in **Regions** put your own first. Colour the ones that are awake right now in green, and enjoy feeling like mission control.

FAMILY ABROAD

Add each family member's city and rename it to the person: **MUM**, **JAMES**, **GRANDAD**. Give them the region **Family** and a colour each. You'll never ring anyone at 3 a.m. again. Probably.

THE SURPRISE

Refresh **At random**, **Between** 5 **and** 25 **minutes apart**, Sound on. Show on all desktops. Every so often the clock clatters into life all by itself, and everybody looks up, just like they used to.

THE EARLY FLIGHT

An alarm at 06:45 on **Weekdays**, labelled WAKE UP, with the chime volume up. Another at 08:30, labelled LEAVE NOW. The board pages you like a passenger who's been in the duty-free too long.

THE CALL TO TOKYO

Add Asia/Tokyo as a place, then an alarm at 09:00, **Time at** TOKYO, labelled CALL TOKYO, on Monday to Friday. It rings when Tokyo's day starts, whatever your clocks are doing for summer time.

THE LAST FLIGHT OUT

Lighting **Late evening**, bulbs **Fail about once a day**, **Replace by hand**, Card brightness halfway up. Dim the room. When a lamp goes, leave it a while: the hall feels emptier for it.

THE QUIET LIBRARY

Refresh **Never**, Sound off, Seconds off, **12-hour**. The cards still flip, silently, once a minute. Very civilised.

GATE 16 WHEN THINGS GO WRONG

Windows won't open it ("Windows protected your PC")

Click **More info**, then **Run anyway**. The clock isn't signed, so SmartScreen doesn't recognise it.

Linux: nothing happens when I open it

Make the file executable first: `chmod +x splitflap-clock-linux-x86_64`. It also needs a desktop with OpenGL, and ALSA for sound, which every ordinary desktop has, PipeWire and PulseAudio ones included.

The time is wrong

The main time is your computer's own clock and time zone. Fix those in your system's date and time settings, and the board follows.

One place is an hour out

Check whether it's **Set by hand**. Hand-set times don't change for summer time. Switch to **From its time zone**, or check you picked the right zone.

No sound

Is **Clatter when the board starts and refreshes** ticked, and the **Volume** up? The clock itself is silent apart from start-ups and refreshes, unless you tick **Also when the time changes**. If Settings shows an orange message, the clock can't find a sound output. It tries again every 15 seconds.

The window has no edges and I can't resize it

That's **Hide title bar**. Use **Size** in Settings, or right-click and untick **Hide title bar**.

I can't find the clock in the taskbar

On Windows, **Show on all desktops** removes its taskbar button. It's still there: look on the desktop and right-click it.

"Show on all desktops" says Wayland doesn't let apps choose this

The clock is running as a native Wayland window, which happens when your session has no X11 (XWayland) at all. Use your desktop's own window menu, often a right-click on the title bar, to put it on all workspaces.

An alarm didn't ring

Is it ticked on, with the right days lit? A **Once** alarm, or one **On a date**, turns itself off after it has rung. Was the clock running? It can't ring while the computer is asleep or the clock is closed, and it only makes up for a missed alarm within 10 minutes of the computer waking. It doesn't ring for a time that had already passed when the clock started or when you set it. Is **Time at** set to the place you meant?

A sky card shows the wrong time of day

Is the place set by hand? Then it goes by its clock, with sunrise and sunset at six. Otherwise it goes by the sun at its time zone's city, which for somewhere far from that city can be a little early or late.

The alarm's row is small

The row fits within the board's width, so the window doesn't jump about when it appears. On a board with only the time on it, that means small cards. Add a place or the date, and it gets bigger.

Part of the board has gone dark

A bulb has died (Gate 10). With **Replace by hand** it stays out until you click the dark lamp at the top, or right-click and choose **Replace the dead bulb**. Bulbs don't fail at all with **Never fail**, or with **Lighting** off.

The cards are hard to make out

Turn up **Card brightness**. If a lighting mood is on, try **Soft lamps** or **Off**.

A place has disappeared

Check its colour: dark grey on dark cards is very discreet. Or it may be further down than you think, under its region's heading.

My settings didn't stick

They're saved when the clock closes normally. If the computer crashed or the clock was killed, it may have missed the last few changes.

GATE 17 THE PART OF TENS

Ten things that are not delays (honest)

1. Names take longer to settle than digits. Their drums have 47 flaps; the digits' have ten at most.
2. A card only ever goes forward. To go from B to A it goes the long way round, through all of C to Z, the digits and the punctuation.
3. When the minute goes from 59 to 00, both cards flip exactly once. The tens drum only has 0 to 5.
4. The hour's tens card is blank before 10 o'clock on a 12-hour clock. Real flap clocks did that.
5. An **é** arrives in a single flip. It isn't on the drum, so it has nothing to come round from.
6. A dark patch on the board is a dead bulb, not a dead clock. The cards under it are still flipping, in the shadows.
7. The start-up is never quite the same twice. The cards set off a moment apart, at their own speeds, on purpose.
8. The clock makes no sound on the minute unless you ask it to. It's a desk clock, not a metronome. The snooze is nine minutes, because every snooze is nine minutes.
9. On Windows, no taskbar button with **Show on all desktops**. That's how Windows does it.
10. The board gets smaller as you add places. The window stops growing at the screen's edge, so the board scales down to fit.

Ten ways to annoy your office with Split-Flap Clock

1. Refresh **At random**, between 1 and 2 minutes apart, with the sound up.
2. Rename everyone's home city to **DELAYED**.
3. Add a place called **LUNCH**, set by hand two hours ahead, and insist it's lunchtime there.
4. An alarm called **BOARDING** at 16:55 every weekday, chime volume all the way up.
5. Give every place a slightly different shade of beige. Call it "1970s accuracy".
6. Bulbs **Fail about once a day**, **Replace by hand**, and tell everyone it's facilities' job.

7. Make a region called **GATE CLOSED** and put your manager's city in it.
8. Announce each refresh over the office in your best airport voice.
9. Add **MARS**, set by hand, and refuse to explain.
10. Choose **Refresh the board** whenever anyone asks you a question, and wait for it to finish before answering.

PHRASEBOOK FOR THE JET-LAGGED

Split-flap display

A display made of drums of hinged cards that fall one by one to show the next symbol. Also called a flap board or a Solari board.

Card, or module

One character's drum of flaps. The clock's main time has four of them, or six with seconds.

Drum

The set of symbols a card carries, in order. Digits have 10, the minute's tens 6, names and dates 47.

Flap

One hinged leaf, carrying the top half of one symbol and the bottom half of the next.

Sky card

The card showing morning, day, evening or night at a place, from its sunrise and sunset.

Lamp

One of the warm lights over the board when a **Lighting** mood is on. Its bulb can flicker, or die.

Snooze

Nine more minutes. Click the board while an alarm rings.

Refresh

Every card spinning once round its drum and landing back where it was.

Region

Your name for a group of places on the board, with a heading of its own.

Time zone

An area that keeps the same clock, named after a big city in it, such as **Europe/Paris**. It knows its own summer time rules.

UTC

Coordinated Universal Time, the world's reference clock, near enough Greenwich Mean Time. Every other time is so many hours and minutes ahead of it or behind it.

Offset

How far a place is ahead of UTC or behind it, as in UTC+05:30. A time set by hand is a fixed offset.

Summer time (DST)

The hour many places move their clocks forward in summer, and back again in autumn. Time zones follow it; hand-set times don't.

Virtual desktop, or workspace

Separate screens' worth of windows on one computer, switched between with a keypress or gesture.

Hex code

A colour written as `#RRGGBB`: red, green and blue, each from 00 to FF.

XWayland

The part of a Wayland desktop that runs older X11 programs. The clock uses it on Linux so that it can stay on top and on every workspace.

ACKNOWLEDGEMENTS

Special thanks to **Oscar**, our most particular passenger. He asked for unlimited places, times set by hand, colours for everything, and regions to sort them into. Then he tested the lot and spotted that you could copy a colour but not paste one. Then he asked for the sky, the reasons, the dates, the white cards, the buttons, the flip speed and the card brightness. Much of version 1 is his doing, and nearly all of 1.1. Every board needs someone who reads it properly.

Beer money. If you like Split-Flap Clock and want to support its developer with some beer funding, send it via PayPal to spoolbeermoney@outlook.com. There's a **Beer money** button at the foot of Settings too. Cheers!



This concludes our tour of the board. Please make sure you have all your belongings with you.

Thank you for flying Split-Flap Clock.